In-Class Exercise: Single-Variable Optimization and Logarithms

ECON 320 - Introduction to Mathematical Economics

Muhebullah Karimzada

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1. Maximization of Profit Function
A firm's total revenue and total cost functions are given by:

$$R(Q) = 80Q - 2Q^2, \quad C(Q) = 20 + 10Q$$

Tasks:

1. Find the profit function $\pi(Q)$.
2. Determine the output level Q^* that maximizes profit.
3. Compute the maximum profit.
4. Verify that the second-order condition for a maximum holds.

2. Utility Maximization Problem
An individual's utility function is:

$$U(x) = 4x - 0.5x^2$$

where x is the number of hours spent on leisure.
Tasks:

1. Determine the value of x that maximizes utility.
2. Check whether it is a maximum using the second derivative test.
3. Interpret the economic meaning of your result.

3. Logarithmic Utility and Diminishing Marginal Utility
Consider the utility function:

$$U(x) = \ln(x)$$

Tasks:

1. Compute $U'(x)$ and $U''(x)$.
2. Show that marginal utility is positive but diminishing.
3. If $x = 10$, find the elasticity of utility with respect to x:

$$E_{Ux} = \frac{dU}{dx} \cdot \frac{x}{U}$$

4. Production Function and Marginal Product
A production function is given by:

$$Q = 10L^{0.5}$$

where Q is output and L is labour.
Tasks:

1. Find the marginal product of labour (MPL).
2. At what level of L does diminishing marginal productivity begin?
3. Suppose the wage rate $w = 2$ and output price $p = 5$. Determine the profit-maximizing L^*.

5. Maximization with Logarithmic Revenue
A monopolist faces the demand function:

$$P(Q) = 100 - 4Q$$

and the total cost function:

$$C(Q) = 20Q + 10$$

Define revenue as $R(Q) = P(Q) \cdot Q$. The firm decides to maximize:

$$\Pi(Q) = \ln[R(Q)] - C(Q)$$

Tasks:

1. Express $R(Q)$ and $\Pi(Q)$ explicitly as functions of Q.
2. Find Q^* that maximizes $\Pi(Q)$.
3. Interpret why taking logs of revenue might change the firm's optimal decision compared to maximizing $R(Q) - C(Q)$.

6. Elasticity and Logarithmic Differentiation
If $y = 20x^{0.8}$, use logarithmic differentiation to find the elasticity of y with respect to x.
Hint: Take the natural logarithm of both sides and differentiate with respect to x.

7. Application of Natural Logarithm Properties
Simplify the following and interpret economically:

$$\ln\left(\frac{x^3y^2}{z^4}\right)$$

If x represents capital, y labour, and z land, explain what happens to the output elasticity if x doubles while y and z remain constant.